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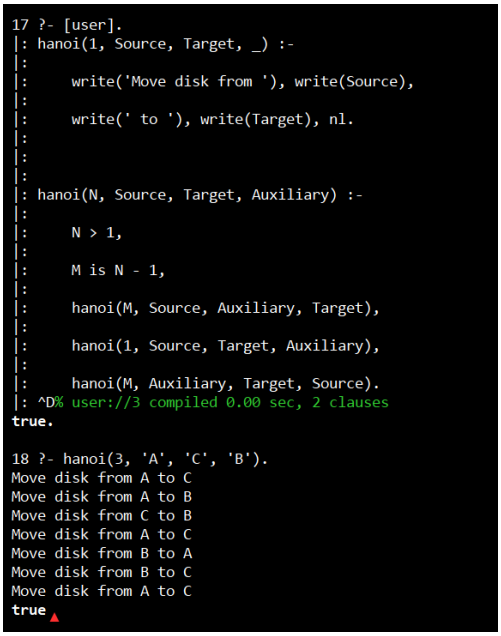
REG.NO:192325037

Ex 21: Towers of Hanoi in Prolog:

PROLOG CODE:

```
hanoi(1, Source, Target, _) :-
   write('Move disk from '), write(Source),
   write(' to '), write(Target), nl.
hanoi(N, Source, Target, Auxiliary) :-
   N > 1,
   M is N - 1,
   hanoi(M, Source, Auxiliary, Target),
   hanoi(1, Source, Target, Auxiliary),
   hanoi(M, Auxiliary, Target, Source).
```

OUTPUT:



```
17 ?- [user].
|: hanoi(1, Source, Target, _) :-
|:
|:     write('Move disk from '), write(Source),
|:
|:     write(' to '), write(Target), nl.
|:
|:
|:
|: hanoi(N, Source, Target, Auxiliary) :-
|:
|:     N > 1,
|:
|:     M is N - 1,
|:
|:     hanoi(M, Source, Auxiliary, Target),
|:
|:     hanoi(1, Source, Target, Auxiliary),
|:
|:     hanoi(M, Auxiliary, Target, Source).
|: ^D% user://3 compiled 0.00 sec, 2 clauses
true.

18 ?- hanoi(3, 'A', 'C', 'B').
Move disk from A to C
Move disk from A to B
Move disk from C to B
Move disk from A to C
Move disk from B to A
Move disk from B to C
Move disk from A to C
true
```